

Block 1 ex 7

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- 1. State whether the assumptions usually adopted in a Gaussian linear regression model (linearity of the relationship, normality and homoscedasticity of the errors, independence of the observations) are satisfied by the model above.
- 2. The model can be written in matrix form $Y = X\beta + \varepsilon$. Write down the explicit expression of the design matrix X .
- 3. Argue what the distribution of the maximum likelihood estimator of β should be.
- 4. Suppose that the estimates $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}) = (2.86, 0.1, 0.86)$ have been obtained. Derive the exact 0.90 confidence intervals for β_1 and β_2 .
- 5. Which additional information would be needed in order to compute R^2 ?

Consider the following regression model:

$$Y_i = \begin{cases} \beta_1 + \varepsilon_i, & i = 1, \dots, 5 \\ \beta_1 + \beta_2(i - 5) + \varepsilon_i, & i = 6, \dots, 10 \end{cases}$$

where $\varepsilon_i, i = 1, \dots, 10$ are independent random variables with distribution $N(0, \sigma^2)$.

1. State whether the assumptions usually adopted in a Gaussian linear regression model (linearity of the relationship, normality and homoscedasticity of the errors, independence of the observations) are satisfied by the model above.

All four standard assumptions of the Gaussian linear regression model are satisfied: the model is linear in the parameters (piecewise linear in i), the errors ε_i are independent, normally distributed with mean 0 and common variance σ^2 , so linearity, normality, homoscedasticity and independence all hold.

2. The model can be written in matrix form $Y = X\beta + \varepsilon$. Write down the explicit expression of the design matrix X .

The vector of parameters is $\beta = (\beta_1, \beta_2)^\top$.

From the piecewise definition:

- For $i = 1, \dots, 5$: $Y_i = \beta_1 + \varepsilon_i \Rightarrow \text{row}(1, 0)$.
- For $i = 6, \dots, 10$: $Y_i = \beta_1 + \beta_2(i - 5) + \varepsilon_i \Rightarrow \text{row}(1, i - 5)$.

So the matrix X (10×2) is:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix}$$

3. Argue what the distribution of the maximum likelihood estimator of β should be.

Since this is a Gaussian linear model with design matrix X and errors $\varepsilon \sim N(0, \sigma^2 I)$, the maximum likelihood estimator of β is

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Because $Y \sim N(X\beta, \sigma^2 I)$, any linear transformation of Y is still normal; therefore

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^\top X)^{-1}).$$

So the MLE of β has a multivariate normal distribution with mean equal to the true parameter vector β and covariance matrix $\sigma^2 (X^\top X)^{-1}$.

4. Suppose that the estimates $(\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}) = (2.86, 0.1, 0.86)$ have been obtained. Derive the exact 0.90 confidence intervals for β_1 and β_2 .

To obtain the exact 90% confidence intervals we use the usual t-based formula for a Gaussian linear model.

From point 2 we have

$$(X^\top X)^{-1} = \frac{1}{425} \begin{pmatrix} 55 & -15 \\ -15 & 10 \end{pmatrix}.$$

Hence

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \frac{55}{425}, \quad \text{Var}(\hat{\beta}_2) = \sigma^2 \frac{10}{425},$$

so the standard errors are

$$\text{se}(\hat{\beta}_1) = \hat{\sigma} \sqrt{\frac{55}{425}}, \quad \text{se}(\hat{\beta}_2) = \hat{\sigma} \sqrt{\frac{10}{425}},$$

with $\hat{\sigma} = 0.86$ and $n - p = 10 - 2 = 8$ degrees of freedom.

Let $t_{0.95,8}$ be the 95th percentile of a t distribution with 8 degrees of freedom.

The exact 90% confidence intervals are then

$$\beta_1 \in \hat{\beta}_1 \pm t_{0.95,8} \text{se}(\hat{\beta}_1) = 2.86 \pm t_{0.95,8} \hat{\sigma} \sqrt{\frac{55}{425}},$$

$$\beta_2 \in \hat{\beta}_2 \pm t_{0.95,8} \text{se}(\hat{\beta}_2) = 0.10 \pm t_{0.95,8} \hat{\sigma} \sqrt{\frac{10}{425}}.$$

5. Which additional information would be needed in order to compute R^2 ?

To compute R^2 we would additionally need the total sum of squares (or the sample variance and mean of the observed Y_i , i.e. the actual data values), in order to calculate $R^2 = 1 - \text{RSS}/\text{TSS}$.